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الرواد الرقميون DIGILIANS

وزارة الاتصالات وتكنولوجيا المعلومات

Under Supervision of:
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AI-Based Driver Safety and Assistance System(ADAS)

SentryEye

AI-BASED DRIVER SAFETY AND ASSISTANCE SYSTEM

REAL-TIME DROWSINESS DETECTION. SMARTER ALERTS. SAFER ROADS.

A fully on-device, browser-native AI system that detects driver fatigue through eye closure, yawning, PERCLOS, and micro-event analysis.

RESULTS

mAP@50 86.75%	mAP@50 86.75%
Precision 90.23%	Recall 88.38%
F1-Score 89.23%	Inference Speed 97.2 FPS

ABSTRACT

SentryEye is an on-device AI driver safety system that detects eye closure and yawning in real time. Using YOLO26n, PERCLOS, and micro-event analysis, the system interprets fatigue levels and delivers visual and audio alerts directly in the browser without uploading video.

SYSTEM OVERVIEW

Camera / Video Input → Frame Processing → YOLO26n Inference → PERCLOS + Micro-events → Debounced Alert State

HUD + audio alert + safety report follow the debounced state machine. Every node, arrow, and label is editable.

QR CODE

https://sharp-gaze-platform.lovable.app

https://dashboard-grad-project.vercel.app

METHODOLOGY

1. Dataset Construction → Data Cleaning & Label Collection

Dataset Construction · Data Cleaning & Label Collection · Model Training · Cross-Model Evaluation

2. Production Pipeline → Browser Deployment

Model Selection · ONNX Export · WebGPU / WASM Deployment · Temporal Fatigue Analysis · Real-Time Alerting

Production model: YOLO26n at 480px and 960px, exported to ONNX and executed in-browser through WebGPU / WASM.
Only the replaceable icons are decorative; the shapes, arrows, and text all remain editable.

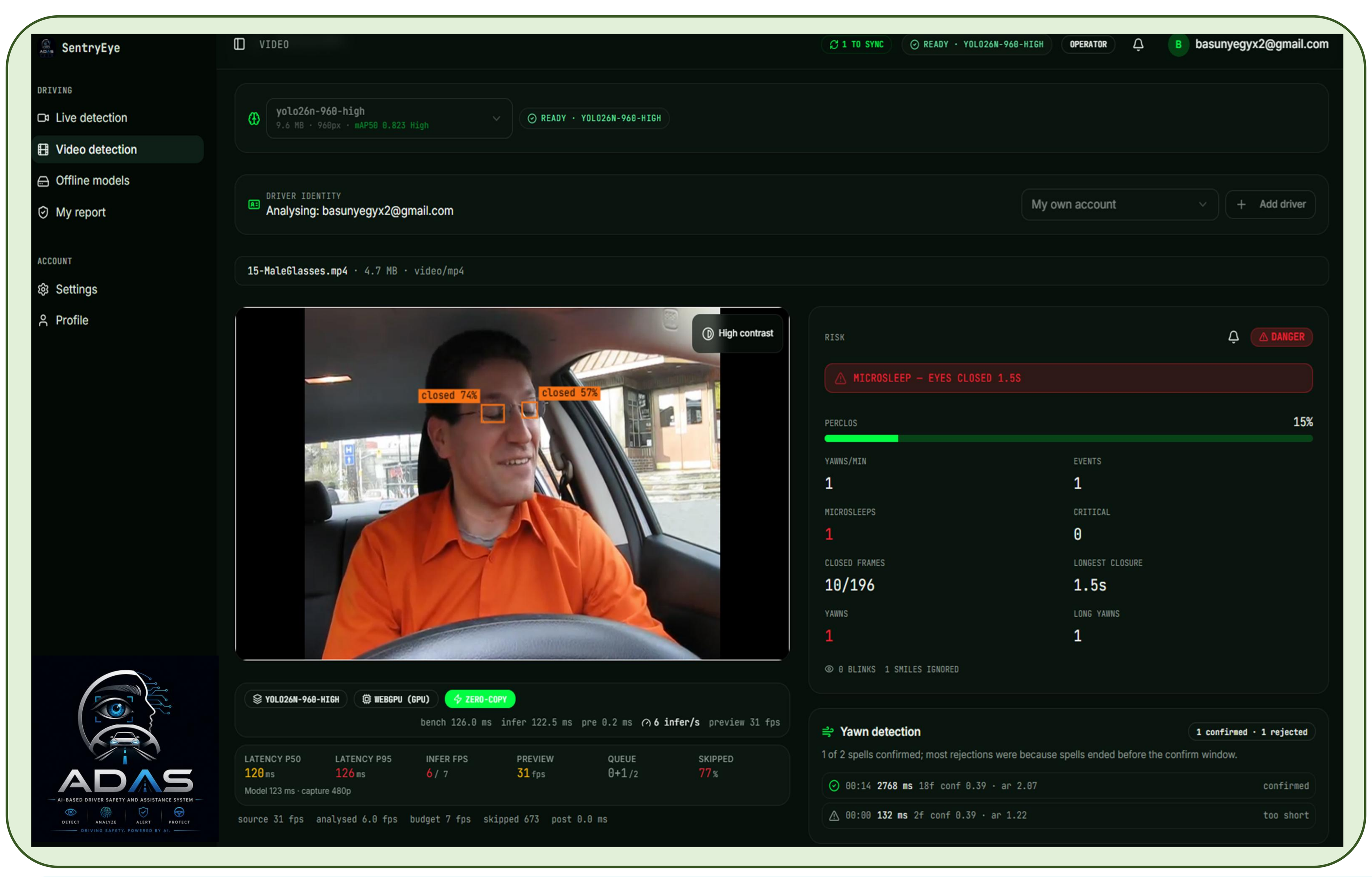
Conclusion

SentryEye demonstrates that real-time driver drowsiness detection can be deployed directly in the browser without transmitting video to a remote server. Visual detection, PERCLOS, and real-time alerts provide a practical foundation for privacy-preserving driver safety.

Aim of the work

To develop a real-time, camera-based driver drowsiness detection system that combines reliable computer vision with temporal fatigue analysis and privacy-preserving browser deployment.

Our Platform



DATASET

50,654

IMAGES · 68,292 ANNOTATED INSTANCES

39,627 TRAIN	5,438 VAL	5,589 TEST

3 CLASSES

- closed_eye
- open_eye
- yawning

Augmentation: HSV jitter, rotation, shear, translation, random erasing (moderate + worst-case tiers).
Mosaic / mixup / copy-paste deliberately disabled.

THE PROBLEM

Driver fatigue is a leading cause of road traffic collisions. Prolonged eye closure and yawning are visible warning signs — but existing detection datasets are too fragmented and inconsistently annotated to build a reliable detector directly.

Offline models

Save a detection model to device once. Live and video detection then start instantly and keep working with no internet connection.

Future Work

- ✓ *Learned temporal modeling (RNN / LSTM) for blink & yawn duration*
- ✓ *Multi-task learning — facial landmarks + gaze*
- ✓ *Multi-camera support*
- ✓ *Context-aware driver personalization*

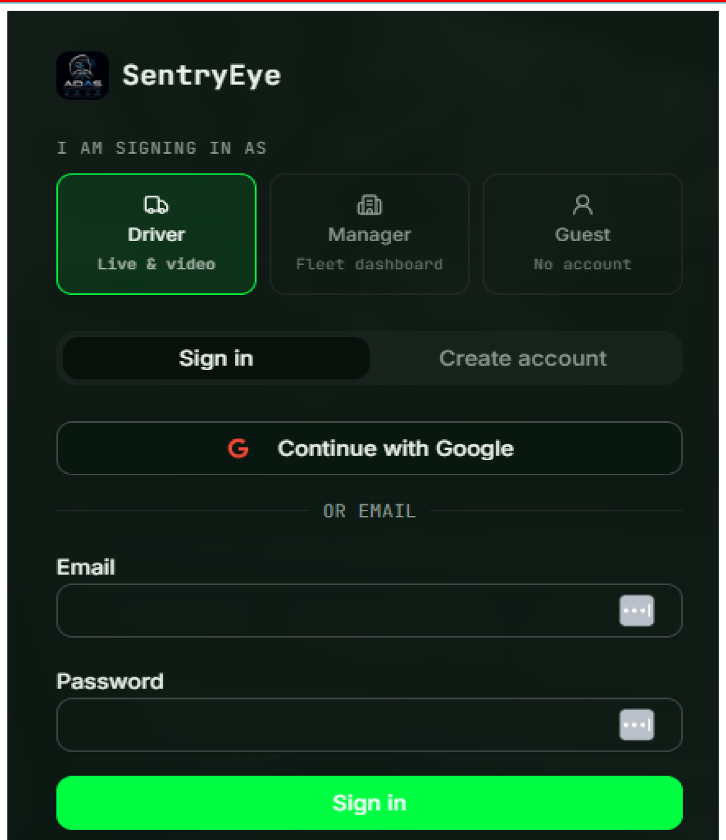
OUR SOLUTION

SentryEye detects three visual states per frame — closed eye, open eye, yawning — then aggregates them into a rolling PERCLOS score and a debounced SAFE / WARNING / CRITICAL alert, so a single misdetection never triggers a false alarm.

The Privacy

Driver Report

The driver will only be able to view his reports status so that he can manage his status to avoid getting fired



Manger report

Managers can view all drivers' aggregated metrics and multi-month performance histories—without live camera feeds—ensuring privacy and informed decision-making

KEY FEATURES

Offline-capable — works fully in-browser without needing constant internet connection

Lightweight, on-device deployment — YOLO11n model compressed below 10MB, runs via WebGPU with no server dependency

AI MODELS

4 architecture families compared under one protocol:
YOLO26 (n / s) · YOLOv11 (n / m) · RF-DETR (Nano / Small) · Faster R-CNN (custom)
21 trained configurations
950 epochs · 277.6 GPU-hours (RTX 2000 Ada)
Production model: YOLO26n – 480px & 960px + ONNX + WebGPU / WASM

USED TOOLS

TensorFlow · Cursor · ONNX · roboflow · supabase · Lovable · PyTorch · kaggle · Claude · Antigravity · OpenAI Codex · Visual Studio Code